

Coding & Solution

Date	29 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

The OptiCrop project implements a machine learning-based crop recommendation system that predicts the most suitable crop using soil nutrients and environmental parameters. The solution is developed using Python, Flask, HTML, CSS, and JavaScript, with a trained Scikit-learn model for prediction. The project follows a modular structure, clean coding practices, and version control using GitHub, making it maintainable, scalable, and easy to deploy.

Instructions:

1. The OptiCrop application was developed using Python, Flask, HTML, CSS, and JavaScript following a modular and well-structured coding approach.
2. The implemented solution addresses the problem of crop recommendation by predicting the most suitable crop using soil nutrients and environmental parameters.
3. The project includes all required source code, trained Machine Learning model, templates, static files, dataset, configuration files, and setup instructions for execution.
4. The complete project source code is maintained in the GitHub repository, and the deployed application is available through the provided project links.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Mohammed-Sadia-Rahmatunnisa/OptiCrop
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn

Key Features Implemented	Crop recommendation using Machine Learning, soil & weather data analysis, user-friendly web interface, trained model prediction
Pending / Incomplete Features	None
Setup / Run Instructions	Install required Python packages, run app.py, and open the Flask application in a web browser.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The OptiCrop project follows a modular architecture using Flask and Machine Learning. The application is well organized, readable, and maintained through GitHub version control. It successfully predicts suitable crops based on soil nutrients and environmental conditions while providing a simple and user-friendly web interface.